

MATH Unit 5 Previous Year Questions

Mathematics (MATH) · Unit 5 · Graph Theory & Combinatorics · 2023

Questions collected from previous year university examinations (2019–2023).

Part A — Short answer (2 marks)

1. Write short notes on Permutations and Combinations. [2022]
2. Illustrate Pigeonhole Principle with a worked example. [2023]
3. Illustrate Recurrence Relations with a worked example. [2020]
4. Explain Euler and Hamiltonian Paths with a neat diagram and a suitable example. [2019]

Part B — Long answer (10 marks)

1. Compare Graph Colouring with Permutations and Combinations. When would you prefer one over the other? [2020]
2. Compare Planar Graphs with Pigeonhole Principle. When would you prefer one over the other? [2020]
3. Illustrate Permutations and Combinations with a worked example. [2020]
4. Discuss the role of Pigeonhole Principle in Graph Theory & Combinatorics. [2020]
5. Discuss the role of Recurrence Relations in Graph Theory & Combinatorics. [2020]
6. Illustrate Euler and Hamiltonian Paths with a worked example. [2020]
7. Compare Graph Colouring with Permutations and Combinations. When would you prefer one over the other? [2019]
8. Compare Planar Graphs with Permutations and Combinations. When would you prefer one over the other? [2022]
9. What is meant by Permutations and Combinations? Discuss its advantages and limitations. [2019]
10. Discuss the role of Pigeonhole Principle in Graph Theory & Combinatorics. [2023]